Adaptive Pathways: data report

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Repository: https://github.com/Muhammadabrarrayhan2/adaptive-pathways-appinventor

Generated by `python run_pipeline.py report`. All numbers below are recomputed from `data/` on every run.

1. Corpus

source	total	in_analysis
github	374	371
official	14	14
gallery	12	11

- Projects stored: **400**; used in analysis: **396** (excluded: 3 structural near-duplicates, 1 empty, 0 parse errors).
- Downloads rejected before storage (3):
- `not_zip` (2): not a ZIP archive (empty or placeholder files committed in place of a project)
- `too_large` (1): archive above the 25 MB size limit
- Licences (top): `{'MIT': 253, 'GPL-3.0': 66, 'Apache-2.0': 21, 'CC0-1.0': 18, 'CC BY-SA 4.0 (MIT App Inventor)': 14, 'CC BY-SA 4.0 (App Inventor Gallery terms)': 12, 'Unlicense': 4, 'GPL-2.0': 4, 'BSD-2-Clause': 2, 'CC-BY-4.0': 1}`

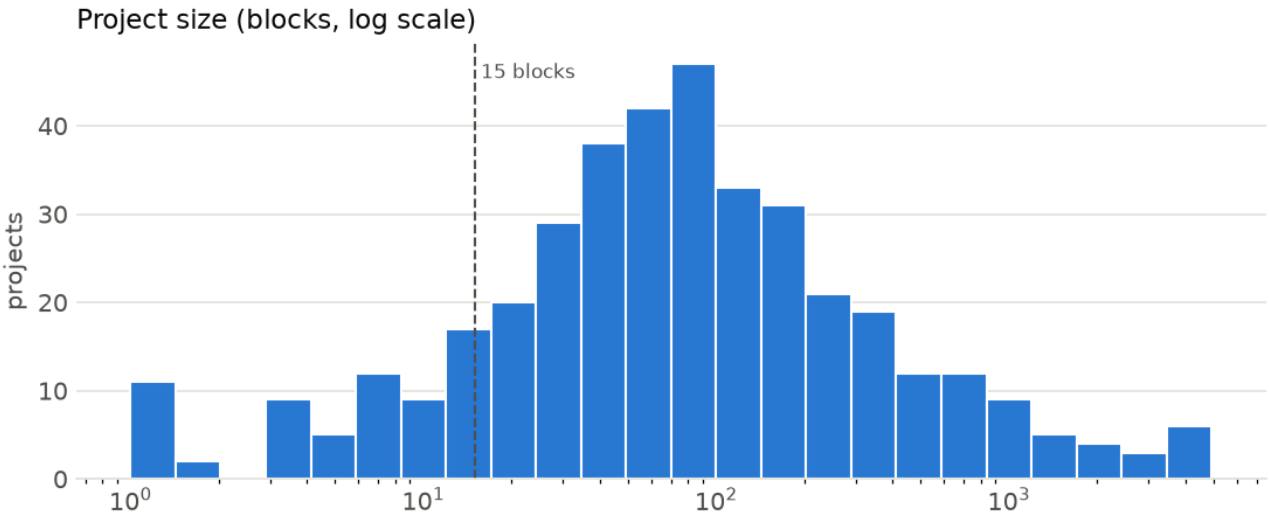
2. Parser quality

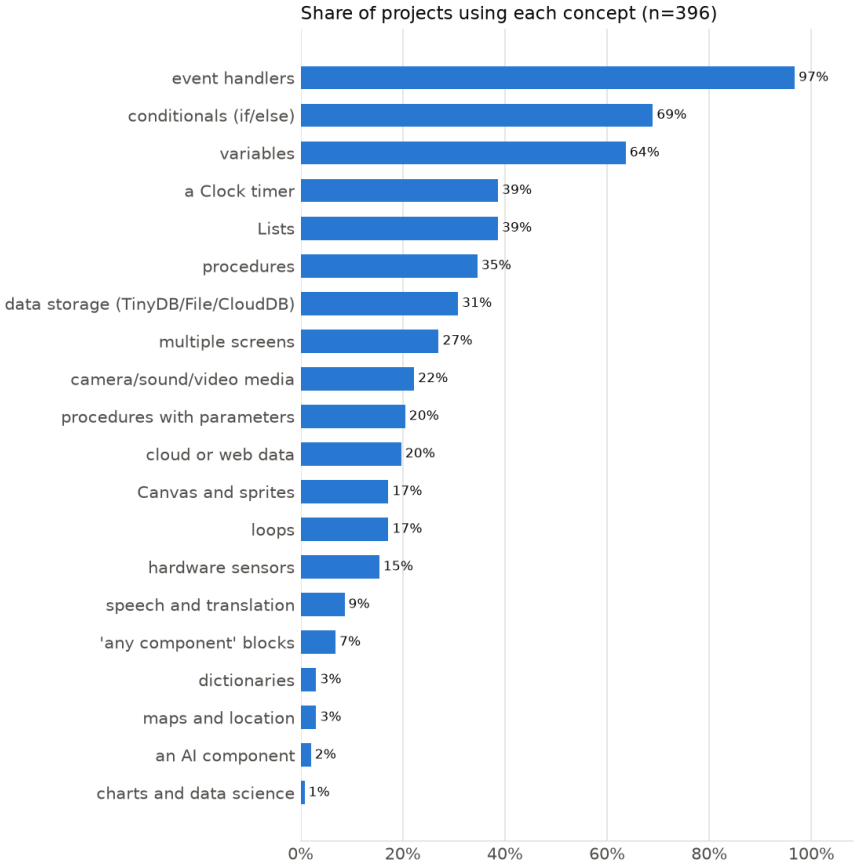
- aiatools parsed **98.5%** of projects; the rest used the direct XML/JSON reader.
- Agreement between aiatools and the direct reader: blocks **99.7%**, components **99.2%**.
- 6 projects use the pre-2013 block format (`Button1_Click`); they are normalised to the modern vocabulary.
- Known aiatools issue handled: project-level `AIAFile.blocks()` drops blocks in multi-screen projects when block ids repeat across screens; the pipeline aggregates `screen.blocks()` per screen instead. Reported upstream as mit-cml/aiatools#13, with a proposed fix in mit-cml/aiatools#14.

3. Comparison with the published Gallery baseline

metric	this corpus (n=396)	Alves, Gresse von Wangenheim & Hauck (2020)
Mean blocks per project	261.6	162.5
Projects with ≤ 15 blocks	15.2%	≥ 25% (first quartile = 15 blocks)
Projects with ≥ 1 event block	96.7%	91.2% (80,832 / 88,606)
Projects with ≥ 1 loop block	17.2%	6.4% (5,635 / 88,606)
Projects with a User Interface component	97.7%	93%

Baseline: Alves, N. d. C., Gresse von Wangenheim, C., & Hauck, J. C. R. (2020). *Teaching Programming to Novices: A Large-scale Analysis of App Inventor Projects.* arXiv:2006.11327.





Differences in definitions. The two columns are not measured identically, so compare them as orders of magnitude:

- *Population.* The paper analyses 88,606 public Gallery projects downloaded in June 2018 with support from the MIT App Inventor team; this corpus has 396 projects from official tutorials, featured Gallery apps and openly licensed GitHub repositories, collected in 2026.
- *Counting method.* The paper counts token frequencies after a lexical analysis of each `.aia`; this pipeline parses the block trees of every screen (aiatools plus a direct XML reader) and normalises pre-2013 block formats.
- *Small projects.* The paper reports only that the first quartile is 15 blocks ("25% of the projects have up to only 15 programming blocks"), which means at least 25% have ≤ 15 blocks; the baseline is

therefore a lower bound.
- *Loops.* The paper counts loop blocks; this pipeline also counts dictionary-walk loops (`controls_for_each_dict`).
- *User Interface.* The paper reports the share of projects with a User Interface palette component (93%); there is no Button- or Label-specific figure in the paper, so this corpus is measured with the same palette category.

Differences are expected: this corpus over-samples curated sources (official tutorials, featured Gallery apps, GitHub repositories), so it skews larger and more structured than the full Gallery.

4. Clustering (RQ1)

Primary solution: **kproto, k = 3** (silhouette 0.133, stability ARI 0.989) — rule: max silhouette over (method, k) with stability ARI >= 0.6 and min cluster size >= 3.

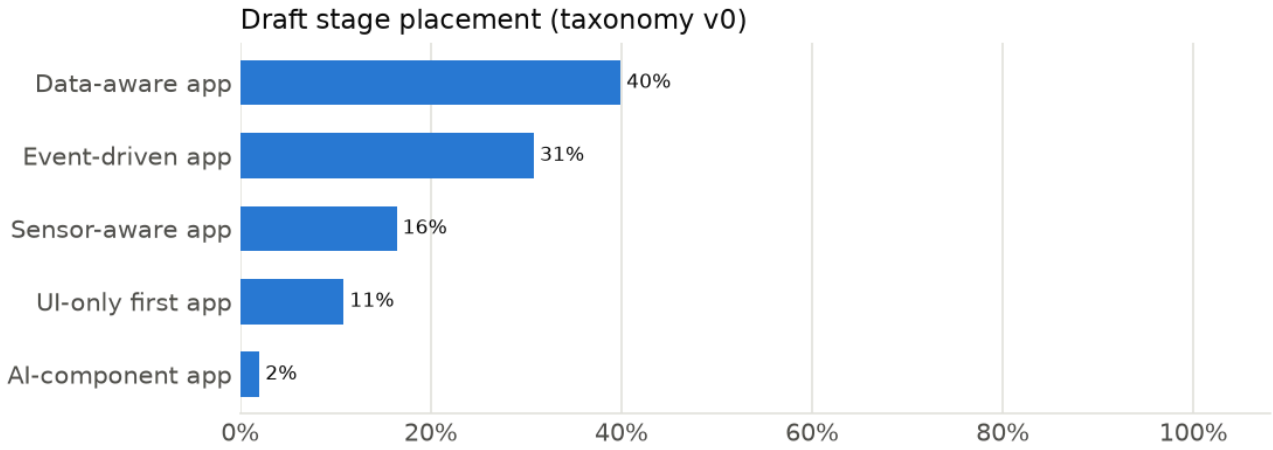
Silhouette values around 0.1 indicate overlapping rather than sharply separated groups: learner projects form a continuum, which is consistent with treating stages as a progression rather than discrete types.

k	silhouette_hier	silhouette_kproto	kproto_seed_stability_ari	hier_bootstrap_stability_ari	hier_vs_kproto_ari	min_cluster_size_hier	min_cluster_size_kproto
3	0.095	0.133	0.989	0.457	0.358	100	81
4	0.107	0.132	0.690	0.370	0.402	59	40
5	0.102	0.114	0.914	0.528	0.362	13	24
6	0.108	0.103	0.647	0.488	0.339	3	25
7	0.119	0.100	0.569	0.497	0.325	3	11
8	0.127	0.072	0.572	0.565	0.268	3	18
9	0.129	0.104	0.558	0.557	0.278	3	11

5. Draft stages vs. data-driven clusters

ARI between the primary clusters and rule-based draft stages: **0.125** (1 = identical partitions, 0 = chance). Low purity clusters are where the teacher review should focus.

cluster	n	majority_stage	majority_stage_name	purity	stage_mix
0	104	1	UI-only first app	0.413	{"ui_first_app": 43, "event_driven": 35, "data_aware": 14, "sensor_aware": 11, "ai_component": 1}
1	81	3	Data-aware app	0.741	{"data_aware": 60, "sensor_aware": 14, "event_driven": 4, "ai_component": 3}
2	211	3	Data-aware app	0.398	{"data_aware": 84, "event_driven": 83, "sensor_aware": 40, "ai_component": 4}



6. Next-step catalog

The catalog holds **56 next-step resources**: 28 official tutorials, 22 AI curriculum units and 6 block-reference pages. For 13 of them, the concepts they introduce are measured from the resource's own solution project; the rest use the index metadata.

target stage	resources
Event-driven app	23
AI-component app	22
Data-aware app	5
Sensor-aware app	5
UI-only first app	1

7. Limitations

- Gallery listings are only partly reachable without login (featured apps); Gallery projects are collected only through public listings. GitHub repositories (open licences only) over-represent classroom and tutorial work.
- Because classroom and tutorial projects are typically finished, guided work, this over-representation is the likely reason the corpus is larger and more structured than the Gallery baseline in section 3 (mean 262 vs 162.5 blocks; 15% vs at least 25% with ≤ 15 blocks).
- Stage labels are a rule-based draft (taxonomy v0) awaiting the two-teacher review in `taxonomy/stage_labels.yaml`.
- No telemetry, no learner outcomes: everything here is structural and descriptive.